

Members: Felix Anthony (M11415812), Ian Joseph Chandra (D11202805)

Customer Service Agent [LLM Project 1]

#1 Intent Parsing

```
You: where is my order 1001?
[Planner] Calling tools: ['order_lookup']
[Tool] order_lookup: Order details found: {'order_id': 1001, 'customer_id': 2, 'product_name': 'Mechanical Key
board', 'status': 'processing', 'order_date': datetime.datetime(2023, 10, 25, 14, 30), 'delivery_date': None}

Agent: Your order with ID 1001, which is a Mechanical Keyboard, is currently in processing status. Unfortunat
ely, there is no estimated delivery date available at this time. If you have any further questions or need assi
stance, please let me know!
```

#2 OrderLookupTool

```
You: check status of order 7890
[Planner] Calling tools: ['order_lookup']
[Tool] order_lookup: Order details found: {'order_id': 7890, 'customer_id': 2, 'product_name': 'USB-C Hub', 's
tatus': 'processing', 'order_date': datetime.datetime(2023, 10, 20, 8, 20), 'delivery_date': datetime.datetime
(2023, 10, 22, 10, 10)}

Agent: The status of your order (ID: 7890) is currently "processing." It was placed on October 20, 2023, and i
s expected to be delivered on October 22, 2023, at 10:10 AM. If you have any further questions or need assista
nce, feel free to ask!
```

#3 CustomerProfileTool

```
You: show my profile
[Planner] Calling tools: ['customer_profile']
[Tool] customer_profile: Customer profile found: {'customer_id': 2, 'name': 'Bob Johnson', 'email': 'bob@examp
le.com', 'created_at': datetime.datetime(2026, 5, 12, 14, 9, 53)}

Agent: Here is your profile information:

- **Name:** Bob Johnson
- **Email:** bob@example.com
- **Account Created On:** May 12, 2026

If you need any further assistance or wish to update your information, please let me know!
```

#4 RefundTool

```
You: refund order 7890
[Planner] Calling tools: ['request_refund']
[Tool] request_refund: Success: Order 7890 status updated to refund_requested.

Agent: Your refund request for order ID 7890 has been successfully submitted. The status of your order is now
updated to "refund requested." If you have any further questions or need assistance, please feel free to ask!
```

#5 ComplaintLoggerTool

```
You: i want to complain about order 7890 is taking too long
[Planner] Calling tools: ['log_complaint']
[Tool] log_complaint: Success: Complaint logged for order 7890.

Agent: Your complaint regarding order ID 7890, stating that it is taking too long to process, has been success
fully logged. If you have any further concerns or need additional assistance, please do not hesitate to reach
out.
```

#6 Multi-step Reasoning

```
You: check status of order 1001, if it is not delivered please refund it!
[Planner] Calling tools: ['order_lookup']
[Tool] order_lookup: Order details found: {'order_id': 1001, 'customer_id': 2, 'product_name': 'Mechanical Key
board', 'status': 'processing', 'order_date': datetime.datetime(2023, 10, 25, 14, 30), 'delivery_date': None}
[Planner] Calling tools: ['request_refund']
[Tool] request_refund: Success: Order 1001 status updated to refund_requested.

Agent: The status of your order (ID: 1001) is currently "processing," and it has not yet been delivered. I hav
e successfully submitted a refund request for this order, and its status is now updated to "refund requested."
If you have any further questions or need assistance, please let me know!
```

#7 Short-Term Memory (STM)

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```
Agent: Your complaint regarding order ID 7890, stating that it is taking too long to process, has been successfully logged. If you have any further concerns or need additional assistance, please do not hesitate to reach out.

You: check status of order 1001, if it is not delivered please refund it!
[Planner] Calling tools: ['order_lookup']
[Tool] order_lookup: Order details found: {'order_id': 1001, 'customer_id': 2, 'product_name': 'Mechanical Keyboard', 'status': 'processing', 'order_date': datetime.datetime(2023, 10, 25, 14, 30), 'delivery_date': None}
[Planner] Calling tools: ['request_refund']
[Tool] request_refund: Success: Order 1001 status updated to refund_requested.

Agent: The status of your order (ID: 1001) is currently "processing," and it has not yet been delivered. I have successfully submitted a refund request for this order, and its status is now updated to "refund requested." If you have any further questions or need assistance, please let me know!

You: what did i refund just now?

Agent: You have just requested a refund for order ID 1001, which contains a "Mechanical Keyboard." If you have any further questions or need assistance, please feel free to ask!
```

#8 Long-Term Memory (Read)

```
felix@ACER MINGW64 ~/projects/llm/customer_service (main)
$ uv run main_new.py

=====
Customer Service Agent (type 'quit' or 'exit' to stop)
=====

You: remember if an order takes too long, refund it!
[Planner] Calling tools: ['store_memory']
[Tool] store_memory: Memory stored: refund_on_delay = if an order takes too long, refund it

Agent: Thank you for your message. I understand that you would like to ensure that refunds are issued if an order takes too long. Please provide me with the details of your order, and I will be happy to assist you with your request.

You: exit
Goodbye!

felix@ACER MINGW64 ~/projects/llm/customer_service (main)
$ uv run main_new.py

=====
Customer Service Agent (type 'quit' or 'exit' to stop)
=====

You: my order 20001 is taking too long!
[Planner] Calling tools: ['order_lookup']
[Tool] order_lookup: Order details found: {'order_id': 20001, 'customer_id': 2, 'product_name': 'Laptop Stand', 'status': 'processing', 'order_date': datetime.datetime(2026, 5, 31, 0, 46, 38), 'delivery_date': None}
[Planner] Calling tools: ['request_refund']
[Tool] request_refund: Success: Order 20001 status updated to refund_requested.

Agent: I understand that your order (ID: 20001) is taking too long. I have initiated a refund for you, and the status has been updated to "refund requested." If you have any further questions or need assistance, feel free to ask!
```

#9 Long-Term Memory (Write)

```
You: remember that if an order is taking too long refund it!
[Planner] Calling tools: ['store_memory']
[Tool] store_memory: Memory stored: refund_policy = if an order is taking too long, refund it

Agent: Thank you for your message. I understand your concern regarding the order delay. If an order is taking longer than expected, we can certainly look into the possibility of a refund. Please provide me with your order details, and I will assist you further.
```

#10 Personalization

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```
Felix@ACER MINGW64 ~/projects/llm/customer_service (main)
$ uv run main_new.py

=====
Customer Service Agent (type 'quit' or 'exit' to stop)
=====

You: my order 1001 came late, remember it!
[Planner] Calling tools: ['store_memory', 'log_complaint']
[Tool] store_memory: Memory stored: late_order = Order 1001 came late.
[Tool] log_complaint: Success: Complaint logged for order 1001.

Agent: I've noted that your order 1001 came late, and I've also logged a complaint regarding this issue. If there's anything else you'd like to address or if you have any questions, please let me know!

You: exit
Goodbye!

Felix@ACER MINGW64 ~/projects/llm/customer_service (main)
$ uv run main_new.py

=====
Customer Service Agent (type 'quit' or 'exit' to stop)
=====

You: my order 20004 came late again!
[Planner] Calling tools: ['log_complaint']
[Tool] log_complaint: Success: Complaint logged for order 20004.

Agent: I have logged your complaint regarding order 20004, which arrived late again. I understand how frustrating this can be, especially since you've experienced this issue before. If there's anything else I can assist you with, please let me know!
```

#11 Verifier

```
You: refund order 123
[Planner] Calling tools: ['request_refund']
[Tool] request_refund: Failed: No order found with ID 123 to update.

Agent: It seems that there is no order found with the ID 123. Could you please double-check the order number? If you have another order ID or need assistance with something else, feel free to let me know!
```